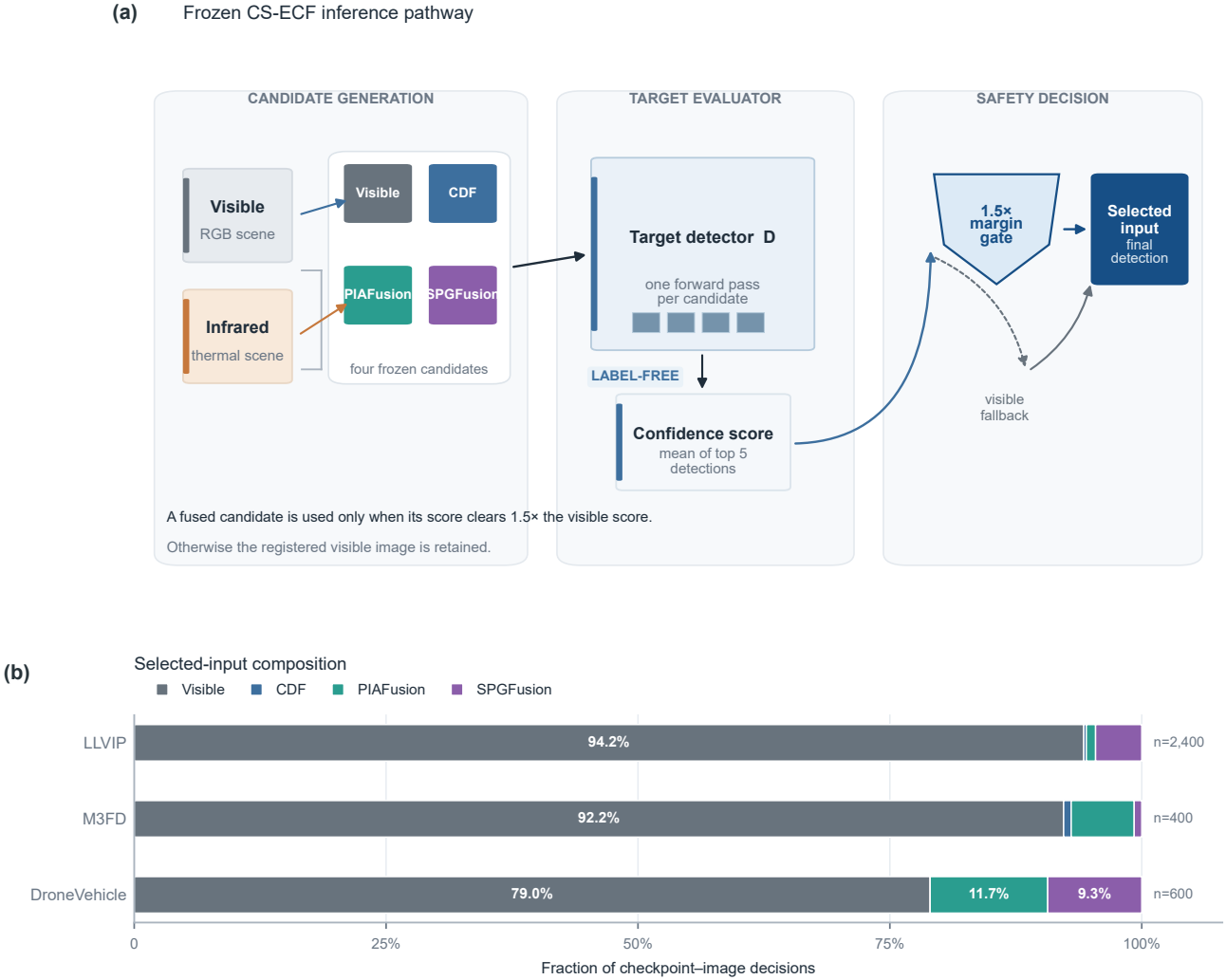

(a)
Frozen CS-ECF inference pathway
CANDIDATE GENERATION
TARGET EVALUATOR
SAFETY DECISION
Visible
RGB scene
Infrared
thermal scene
Visible
CDF
PIAFusion
SPGFusion
four frozen candidates
Target detector D
one forward pass per candidate
LABEL-FREE
Confidence score
mean of top 5 detections
1.5× margin gate
Selected input
final detection
visible fallback
A fused candidate is used only when its score clears 1.5× the visible score.
Otherwise the registered visible image is retained.
(b)
Selected-input composition
Visible
CDF
PIAFusion
SPGFusion
LLVIP
94.2%
n=2,400
M3FD
92.2%
n=400
DroneVehicle
79.0%
11.7%
9.3%
n=600
0
25%
50%
75%
100%
Fraction of checkpoint–image decisions